

## DNA TEST REPORT – MEDGENOME LABORATORIES

|                      |                                                                                                               |                            |                                |
|----------------------|---------------------------------------------------------------------------------------------------------------|----------------------------|--------------------------------|
| Full Name / Ref No:  | Baby of Brindha                                                                                               | Order ID/Sample ID:        | 598094/7864291                 |
| Date of Birth / Age: | 1 Month                                                                                                       | Gender:                    | Male                           |
| Parental Sample ID:  | NA                                                                                                            | Sample Type:               | Peripheral Blood in EDTA       |
| Referring Clinician: | Dr. Kiruthika,<br>Salem Genetics Centre – Salem.                                                              | Date of Sample Collection: | 21 <sup>st</sup> February 2023 |
|                      |                                                                                                               | Date of Sample Receipt:    | 22 <sup>nd</sup> February 2023 |
|                      |                                                                                                               | Date of Order:             | 25 <sup>th</sup> February 2023 |
|                      |                                                                                                               | Date of Report:            | 21 <sup>st</sup> March 2023    |
| Test Requested:      | Congenital adrenal hyperplasia <i>CYP21A2</i> (21-0H) combo - NGS and deletion/duplication analysis [MGM1019] |                            |                                |

## CLINICAL DIAGNOSIS / SYMPTOMS / HISTORY

The patient, born of a consanguineous marriage, presented with Pulmonary arterial hypertension (PAH), hypospadias, undescended unilateral testis, accessory nipple, and sacral dimple. The patient elder sibling presented with lethargy, poor feeding, respiratory distress syndrome (RDS), undetermined sex (inter sex) and expired at day 7 of life. The patient is suspected to be affected with congenital adrenal hyperplasia (CAH) / Differences in sex development (DSD) / inborn errors of metabolism (IEMs) and is being evaluated for pathogenic variation in *CYP21A2* gene by NGS and MLPA.

## RESULTS\*

**PATHOGENIC VARIANT CAUSATIVE OF THE SUSPECTED PHENOTYPE WAS IDENTIFIED**

| Gene (Transcript) #                              | Location | Variant                             | Zygosity   | Disease(OMIM)                                                     | Inheritance         | Classification |
|--------------------------------------------------|----------|-------------------------------------|------------|-------------------------------------------------------------------|---------------------|----------------|
| <b><i>CYP21A2</i> (+)</b><br>(ENST00000418967.2) | Intron 2 | <b>c.293-13C&gt;G</b><br>(Intronic) | Homozygous | Adrenal hyperplasia, congenital, due to 21-hydroxylase deficiency | Autosomal recessive | Pathogenic     |

\*Genetic test results are reported based on the recommendations of American College of Medical Genetics [1].

## ADDITIONAL FINDINGS: NO VARIANT(S) OF UNCERTAIN SIGNIFICANCE (VUS) IDENTIFIED

No other variant that warrants to be reported was detected.

The *CYP21A2* gene is 100% covered in this assay including deletions/duplications.

## VARIANT INTERPRETATION AND CLINICAL CORRELATION

### NGS

A homozygous splice site variation in Intron 2 of the *CYP21A2* gene (**chr6:g.32006858C>G; Depth: 1002x**) that results in cryptic splicing (**c.293-13C>G; ENST00000418967.2**) was detected in this subject. This variation, commonly known as IVS2-13A/C>G, responsible for loss of functional phenotype and is the most frequent non- deletional variation reported to be

associated with classic congenital adrenal hyperplasia arising from 21-hydroxylase deficiency. The variant has a minor allele frequency of 0.09984%, 0.22000% and 0.20861% in the 1000 genomes, gnomAD and our internal databases respectively. The reference base is conserved across mammals [23-26].

## MLPA

The same homozygous variation (c.293-13C>G) was detected within the detection limits of MLPA, in the *CYP21A2* gene of this subject.

**OMIM phenotype:** Autosomal recessive Congenital Adrenal Hyperplasia (OMIM#201910) is caused by homozygous or compound heterozygous variations in the *CYP21A2* gene (OMIM\*613815).

Based on the above evidences \$, **the variation detected in the *CYP21A2* gene is classified pathogenic and have to be carefully correlated with the clinical symptoms observed.**

## RECOMMENDATIONS

Genetic counselling is advised.

Testing of parents sample is recommended to confirm the carrier status of the detected variants.

If results obtained do not match the clinical findings, additional testing should be considered as per referring clinician's recommendations.

## LIMITATIONS

- Genetic testing is an important part of the diagnostic process. However, genetic tests may not always give a definitive answer. In some cases, testing may not identify a genetic variant even though one exists. This may be due to limitations in current medical knowledge or testing technology. Accurate interpretation of test results may require knowing the true biological relationships in a family. Failing to accurately state the biological relationships in {my/my child's} family may result in incorrect interpretation of results, incorrect diagnoses, and/or inconclusive test results.
- Test results are interpreted in the context of clinical findings, family history and other laboratory data. Only variations in genes potentially related to the proband's medical condition are reported. Rare polymorphisms may lead to false negative or positive results. Misinterpretation of results may occur if the information provided is inaccurate or incomplete.
- Specific events like copy number variations, translocations, repeat expansions and chromosomal rearrangements may not be reliably detected with targeted sequencing. Variants in untranslated region, promoters and intronic variants are not assessed using this method.
- Genetic testing is highly accurate. Rarely, inaccurate results may occur for various reasons. These reasons include, but are not limited to: mislabeled samples, inaccurate reporting of clinical/medical information, rare technical errors or unusual circumstances such as bone marrow transplantation, blood transfusion; or the presence of change(s) in such a small percentage of cells that may not be detectable by the test (mosaicism).

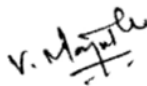
## DISCLAIMER

- Interpretation of variants in this report is performed to the best knowledge of the laboratory based on the information available at the time of reporting. The classification of variants can change over time and MedGenome cannot be held responsible for this. Please feel free to contact MedGenome Labs (techsupport@medgenome.com) in the future to determine if there have been any changes in the classification of any variations. Re-analysis of variants in previously issued reports in light of new evidence is not routinely performed but may be available upon request.
- Genes with pseudogenes, paralog genes and genes with low complexity may have decreased sensitivity and specificity of variant detection and interpretation due to inability of the data and analysis tools to unambiguously determine the origin of the sequence data in such regions.
- The mutations have not been validated/confirmed by Sanger sequencing.
- Incidental or secondary findings (if any) that meet the ACMG guidelines [2] can be given upon request.
- The report shall be generated within turnaround time (TAT), however, such TAT may vary depending upon the complexity of test(s) requested. MedGenome under no circumstances will be liable for any delay beyond afore mentioned TAT.
- It is hereby clarified that the report(s) generated from the test(s) do not provide any diagnosis or opinion or recommends any cure in any manner. MedGenome hereby recommends the patient and/or the guardians of the patients, as the case may be, to take assistance of the clinician or a certified physician or doctor, to interpret the report(s) thus generated. MedGenome hereby disclaims all liability arising in connection with the report(s).
- In a very few cases genetic test may not show the correct results, e.g. because of the quality of the material provided to MedGenome. In case where any test provided by MedGenome fails for unforeseeable or unknown reasons that cannot be influenced by MedGenome in advance, MedGenome shall not be responsible for the incomplete, potentially misleading or even wrong result of any testing if such could not be recognised by MedGenome in advance.
- This is a laboratory developed test and the development and the performance characteristics of this test was determined by MedGenome.



**Jeevana Praharsha.A**

Senior Genome Analyst



**Manjunath V**

Senior Manager (Lab Operations)



**Dr. Vasundhara Parag Tamhankar**

DNB (Paediatrics), DCH, MBBS, Fellowship in  
Genetics, Consultant Clinical Geneticist

## TEST METHODOLOGY

### Next generation sequencing

Targeted gene sequencing: Selective amplification and sequencing of the targeted region of the genome/genes is performed. DNA was used to perform targeted amplification of the complete *CYP21A2* gene. Libraries were prepared using the amplified product and sequenced to mean coverage of >100x on Illumina sequencing platform. Targeted sequencing represents a cost-effective approach to detect variants present in multiple/large genes in an individual.

The sequences obtained are aligned to *CYP21A2* gene sequence only, extracted from human reference genome (GRCh37/hg19) using BWA program [2, 3] and analyzed using Picard and Sentieon version 201808.01 to identify variants relevant to the clinical indication. We follow the GATK best practices framework for identification of variants in the sample [4, 5]. Gene annotation of the variants is performed using VEP program [6] against the Ensembl release 87 human gene model [7]. Clinically relevant mutations were annotated using published variants in literature and a set of diseases databases - ClinVar, OMIM, GWAS, HGMD and SwissVar [8-15]. Common variants are filtered based on allele frequency in 1000Genome Phase 3, ExAC, EVS, dbSNP147, 1000 Japanese Genome and our internal Indian population database [16-20]. Non-synonymous variants effect is calculated using multiple algorithms such as PolyPhen-2, SIFT, Mutation Taster2, Mutation Assessor, and LRT. Only non-synonymous variants, splice site variants and previously reported intronic, silent and promoter variants found in the *CYP21A2* gene were used for clinical interpretation.

### Multiplex Ligation dependent Probe Amplification (MLPA)

Copy number changes in targeted regions of the *CYP21A2* gene were identified by hybridizing with MLPA (Multiplex Ligation dependent Probe Amplification) probes. Each MLPA probe consists of two hemi-probes that bind to adjacent sites on the target sequence. Upon ligation and subsequent PCR amplification, each distinct MLPA probe (specific to distinct target regions) generates an amplicon with a unique length which are separated and quantified by capillary electrophoresis. Heterozygous deletion/SNVs within target sequences will prevent efficient probe binding and give a 35-50% reduced relative peak area of the amplification product specific to that probe set. Copy number differences of various exons between test and control DNA samples can be detected by analyzing the MLPA peak patterns.

**§The classification of the variations is done based on American College of Medical Genetics as described below [1]**

|                                   |                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Variant                           | A change in a gene. This could be disease causing (pathogenic) or not disease causing (benign).                                                                                                                                                                                                                                                                                                                |
| Pathogenic                        | A disease causing variation in a gene which can explain the patients' symptoms has been detected. This usually means that a suspected disorder for which testing had been requested has been confirmed.                                                                                                                                                                                                        |
| Likely Pathogenic                 | A variant which is very likely to contribute to the development of disease however, the scientific evidence is currently insufficient to prove this conclusively. Additional evidence is expected to confirm this assertion of pathogenicity.                                                                                                                                                                  |
| Variant of Uncertain Significance | A variant has been detected, but it is difficult to classify it as either pathogenic (disease causing) or benign (non-disease causing) based on current available scientific evidence. Further testing of the patient or family members as recommended by your clinician may be needed. It is probable that their significance can be assessed only with time, subject to availability of scientific evidence. |

\*The transcript used for clinical reporting generally represents the canonical transcript (according to Ensembl release 87 gene model), which is usually the longest coding transcript with strong/multiple supporting evidence. However, clinically relevant variants annotated in alternate complete coding transcripts could also be reported.

Variants annotated on incomplete and nonsense mediated decay transcripts will not be reported.

\*The *in silico* predictions are based on Variant Effect Predictor, Ensembl release 87 (SIFT version - 5.2.2; PolyPhen - 2.2.2); LRT version - November, 2009 release from dbNSFPv3.1 and Mutation Taster2 based on build NCBI 37 / Ensembl 69 [21].

For any further technical queries please contact [techsupport@medgenome.com](mailto:techsupport@medgenome.com).

## REFERENCES

1. Richards S, Aziz N, Bale S, *et al*, Standards and guidelines for the interpretation of sequence variants: a joint consensus recommendation of the American College of Medical Genetics and Genomics and the Association for Molecular Pathology, Genetics in Medicine, 2015 May;17(5):405-24.
2. Li, H. and R. Durbin. Fast and accurate long-read alignment with Burrows-Wheeler transform. Bioinformatics, 2010. 26(5): p. 589-95.
3. Meyer, L.R., *et al*, The UCSC Genome Browser database: extensions and updates 2013. Nucleic Acids Res, 2013. 41(D1): p. D64-9.
4. McKenna, A., *et al*, The Genome Analysis Toolkit: a MapReduce framework for analyzing next-generation DNA sequencing data. Genome Res, 2010. 20(9): p. 1297-303.
5. Li, H., *et al*, The Sequence Alignment/Map format and SAMtools. Bioinformatics, 2009. 25(16): p. 2078-9.
6. McLaren, W., *et al*, Deriving the consequences of genomic variants with the Ensembl API and SNP Effect Predictor. Bioinformatics, 2010. 26(16): p. 2069-70.
7. ENSEMBL: <http://www.ensembl.org>
8. Landrum, M J., *et al*, ClinVar: public archive of interpretations of clinically relevant variants. Nucleic Acids Res., 2015.
9. Hamosh A., *et al*, Online Mendelian Inheritance in Man (OMIM), a knowledgebase of human genes and genetic disorders, 2005.33:D514-7.
10. OMIM: <http://www.omim.org>
11. GWAS: <http://www.ebi.ac.uk/gwas/>
12. Welter D., *et al*, The NHGRI GWAS Catalog, a curated resource of SNP-trait associations. Nucleic Acids Res., 2014.
13. Stenson, P. D., *et al*, Human Gene Mutation Database (HGMDR): 2003 update. Hum. Mutat. 21, 577-581.
14. HGMD: <http://www.biobase-international.com/product/hgmd>
15. Mottaz, A., *et al*, Easy retrieval of single amino-acid polymorphisms and phenotype information using SwissVar. Bioinformatics, 2010.26(6): p. 851-2.
16. A global reference for human genetic variation. Nature, 2015. 526(7571): p. 68-74.
17. Analysis of protein-coding genetic variation in 60,706 humans. 2015, <http://dx.doi.org/10.1101/030338>
18. NHLBI: <https://esp.gs.washington.edu/drupal>
19. Nagasaki, M., Rare variant discovery by deep whole-genome sequencing of 1,070 Japanese individuals. Nat. Commun. 2015.
20. dbSNP: <http://www.ncbi.nlm.nih.gov/SNP/>
21. Mutation Taster2: <http://www.mutationtaster.org/>
22. Green R. C., *et al*, American College of Medical Genetics and Genomics. ACMG recommendations for reporting of incidental findings in clinical exome and genome sequencing. Genet Med. 2013 Jul;15(7):565-74.
23. Speiser PW, Azziz R, Baskin LS, Ghizzoni L, Hensle TW, Merke DP, *et al*. Congenital adrenal hyperplasia due to steroid 21-hydroxylase deficiency: an Endocrine Society clinical practice guideline. J Clin Endocrinol Metab. 2010 Sep;95(9):4133-60.
24. Marumudi E, Sharma A, Kulshreshtha B, Khadgawat R, Khurana ML, Ammini AC. Molecular genetic analysis of CYP21A2 gene in patients with congenital adrenal hyperplasia. Indian J Endocrinol Metab. 2012 May;16(3):384-8.

25. Higashi Y, Tanae A, Inoue H, Fujii-Kuriyama Y. Evidence for frequent gene conversion in the steroid 21-hydroxylase P-450(C21) gene: implications for steroid 21-hydroxylase deficiency. Am J Hum Genet. 1988 Jan;42(1):17-25.
26. Baumgartner-Parzer S, Witsch-Baumgartner M, Hoepfner W. EMQN best practice guidelines for molecular genetic testing and reporting of 21-hydroxylase deficiency. Eur J Hum Genet. 2020 10;28(10):1341-67.

----- End of Report -----